

1. Administrative & Study Identification

Full Study Title: A Study of Calderasib (MK-1084) in Participants With Hepatic Impairment and Healthy Volunteers (MK-1084-017)
Official Regulatory Title: A Clinical Study to Evaluate the Effect of Hepatic Impairment on the Single-Dose Pharmacokinetics of MK-1084
Short Title/Acronym: MK-1084-017
Protocol Number: MK-1084-017
Other Identifiers: 1084-017
NCT Number: NCT07219550
Sponsor Name: Merck Sharp & Dohme LLC
Investigational Product: Calderasib (MK-1084)
Phase of Development: Phase 1
Overall Status: Recruiting
Medical Monitor: Merck Sharp & Dohme LLC
Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To determine the pharmacokinetic (PK) parameters of calderasib in plasma, including area under the curve (\$AUC\$), maximum observed concentration (\$C_{\text{max}}\$), time to maximum concentration (\$T_{\text{max}}\$), terminal half-life (\$t_{1/2}\$), apparent clearance (\$CL/F\$), and apparent volume of distribution during the terminal phase (\$V_z/F\$).
Secondary Objectives: To evaluate the safety and tolerability of a single oral dose of calderasib (MK-1084) in participants with hepatic impairment and healthy volunteers, including monitoring adverse events (AEs), and to learn what happens to calderasib levels in the body over time.

2.2 Background & Rationale This study is conducted to understand how hepatic (liver) impairment affects the pharmacokinetics, absorption, and safety of calderasib compared to healthy participants. Generating this PK and safety profile is essential for establishing appropriate dosing guidelines for patients with varying degrees of liver dysfunction.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional
Primary Purpose: Basic Science
Interventional Model: Single Group Assignment
Allocation: N/A
Blinding/Masking: None (Open Label)
Control Type: Healthy Volunteer Comparator (Control Group)

3.2 Planned Enrollment & Duration
Target Enrollment (\$N\$): 58 participants
Number of Sites: Multicenter (including sites such as San Antonio, TX and Chandler, AZ)
Actual Study Start: 05-Nov-2025
Estimated Primary Completion: 07-Aug-2026
Estimated Study Completion: 14-Aug-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Sex: All. 2. Age 18 to 80 years. 3. Body mass index (BMI) between 18.0 and 42.0 kg/m^2 . 4. Conditions: Must either have documented hepatic impairment or be a healthy volunteer (medically healthy with no clinically significant medical history).

4.2 Exclusion Criteria 1. History of gastrointestinal disease which may affect food and drug absorption, or a history of cancer (malignancy). 2. Positive result for human immunodeficiency virus (HIV), Hepatitis B surface antigen (HBsAg), or Hepatitis C virus (HCV). 3. For Hepatic Impairment (HI) participants: Severe complications of liver disease within 3 months, history of recent variceal bleeds, hepatorenal syndrome, life expectancy < 3 months, or history of solid organ transplantation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A single oral dose of calderasib on Day 1. **Route of Administration:** Oral Tablet. **Duration of Treatment:** Single-dose administration followed by a pharmacokinetic sampling and safety monitoring period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard stable therapies for existing non-excluded comorbidities. **Prohibited:** HIV protease inhibitors, antiviral and/or immune modulating therapy for HBV or HCV within 90 days prior to dosing.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, BMI check, Organ function grading
Baseline	Day 1	Single Oral Dose of calderasib, Baseline Vitals, Safety audit
Interim	Post-dose Days	Frequent Blood Sampling for PK ($AUC_{0\text{-last}}$, $AUC_{0\text{-inf}}$, $AUC_{0\text{-24}}$, C_{max} , T_{max} , $t_{1/2}$, CL/F , V_z/F), AE Monitoring
End of Study	Final Follow-up	Final Vital Signs, Safety Review, Resolution of AEs

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints - Area under the concentration versus time curve from 0 to the last quantifiable concentration ($AUC_{0\text{-last}}$) of calderasib in plasma. - Area under the concentration versus time curve from 0 to infinity ($AUC_{0\text{-inf}}$) of calderasib in plasma. - Area under the concentration versus time

curve from 0 to 24 hours (AUC_{0-24}) of calderasib in plasma. - Maximum observed concentration (C_{max}) of calderasib in plasma. - Time to maximum concentration (T_{max}) of calderasib in plasma. - Terminal half-life ($t_{1/2}$) of calderasib in plasma. - Apparent clearance (CL/F) of calderasib in plasma. - Apparent volume of distribution during the terminal phase (V_z/F) of calderasib.

7.2 Secondary & Exploratory Endpoints - Number of participants who experience an adverse event (AE). - Number of participants who discontinue the study due to an adverse event (AE).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of all participants; an AE is defined as any untoward medical occurrence associated with the drug, whether drug-related or not. **Serious Adverse Events (SAEs):** Subject to standard 24-hour reporting protocols. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by the sponsor.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving at least one dose) and PK Set (participants with sufficient plasma samples for analysis). **Sample Size Justification:** $N = 58$ participants to adequately represent differing classes of hepatic impairment against a matched cohort of healthy volunteers. **Handling of Missing Data:** Standard non-compartmental analysis (NCA) procedures will be used for missing PK parameters.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and centralized monitoring by the sponsor. **Audit Trail:** Detailed source document verification and electronic Case Report Form (eCRF) tracking.

11. Study Identifiers & Governance

Primary ID: MK-1084-017 **Other Identifiers:** 1084-017 **Registry Number:** NCT07219550 **Sponsor/Lead Organization:** Merck Sharp & Dohme LLC **Study Title:** A Study of Calderasib (MK-1084) in Participants With Hepatic Impairment and Healthy Volunteers (MK-1084-017) **Official Regulatory Title:** A Clinical Study to Evaluate the Effect of Hepatic Impairment on the Single-Dose Pharmacokinetics of MK-1084.

12. Clinical Framework (Hepatic Impairment Specific)

Primary Condition: Hepatic Impairment, Healthy **Diagnosis Threshold:** Clinically diagnosed liver impairment grading compared against medically healthy volunteers. **Medical Methodology:** Pharmacokinetic and Safety Phase 1 trial. **Optimization Target:** Establishment of safe dosing parameters in subjects with altered drug metabolism/clearance.

13. Study Procedures & Methodology

Management Pathway: Intensive PK blood draw schedule following Day 1 administration. **Education Protocol (HCP):** Site initiation and PK protocol adherence training. **Education Protocol (Patient):** Informed consent detailing blood draw frequency and lifestyle restrictions. **Quality Audit Mechanism:** Clinical monitoring of blood draw timing and sample preservation.

14. Observation & Follow-Up Schedule

Total Duration: Time frame requisite to capture full terminal elimination half-life profile of calderasib. **Onsite Visit Cadence:** Confinement or frequent outpatient visits during the active PK sampling window. **Remote Monitoring Frequency:** N/A for Phase 1 PK. **Checklist Review Interval:** Daily during the inpatient/frequent outpatient sampling phase.

15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	Medical Director	_____	03-Apr-2026
Clinical Operations Lead	[Name]	_____	03-Apr-2026
Lead Statistician	[Name]	_____	03-Apr-2026